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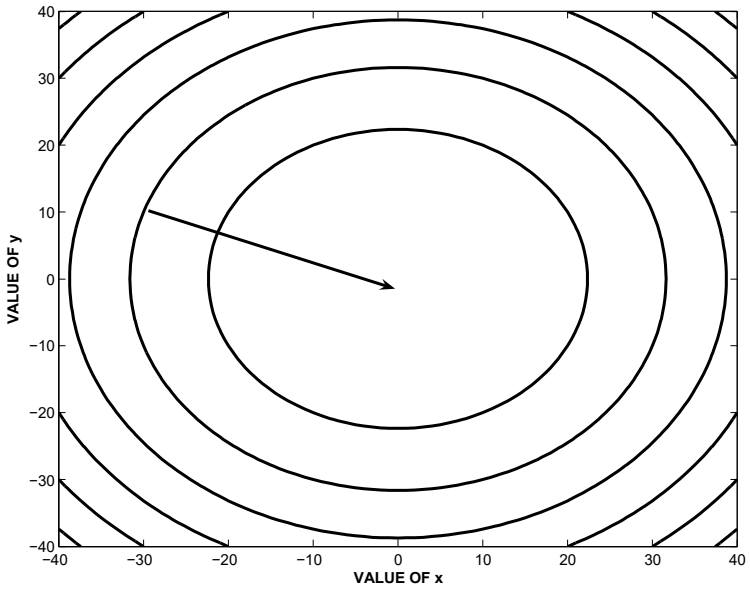
Gradient Ratios, Vanishing and Exploding Gradient Problems

Neural Networks and Deep Learning, Springer, 2018
Chapter 3, Section 3.4

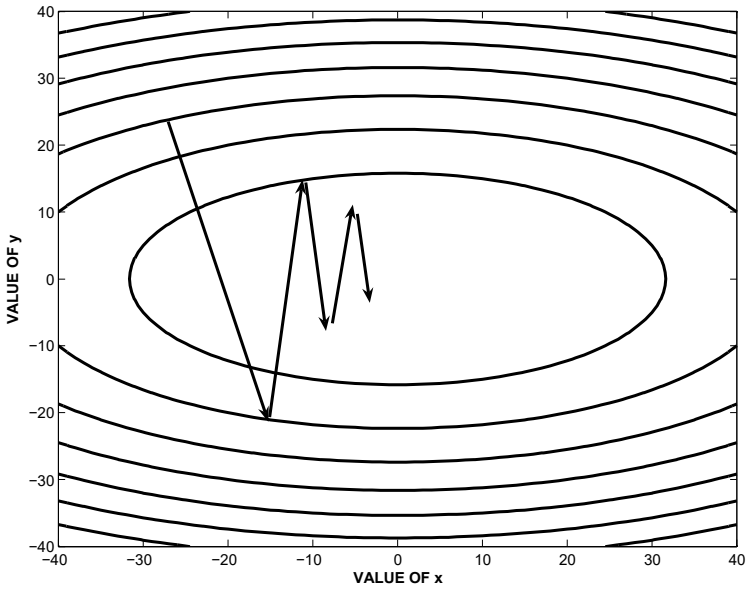
Effect of Varying Slopes in Gradient Descent

- Neural network learning is a *multivariable* optimization problem.
- Different weights have different magnitudes of partial derivatives.
- Widely varying magnitudes of partial derivatives affect the learning.
- Gradient descent works best when the different weights have derivatives of similar magnitude.
  - *The path of steepest descent in most loss functions is only an instantaneous direction of best movement, and is not the correct direction of descent in the longer term.*

Example



(a) Loss function is circular bowl $L = x^2 + y^2$



(b) Loss function is elliptical bowl $L = x^2 + 4y^2$

- Loss functions with varying sensitivity to different attributes

Revisiting Feature Normalization

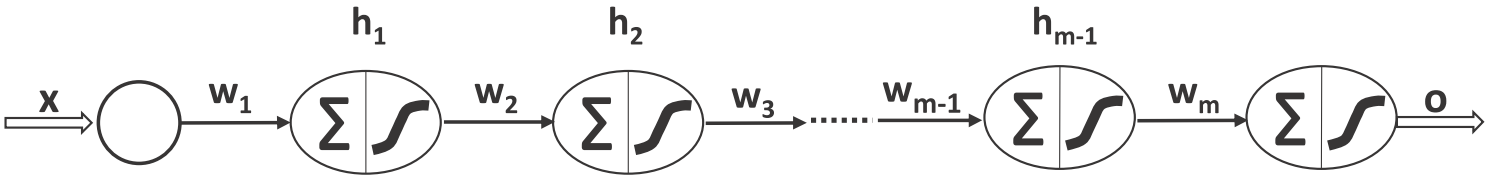
- In the previous lecture, we discussed feature normalization.
- When features have very different magnitudes, gradient ratios of different weights are likely very different.
- Feature normalization helps even out gradient ratios to some extent.
  - Exact behavior depends on target variable and loss function.

The Vanishing and Exploding Gradient Problems

- An extreme manifestation of varying sensitivity occurs in deep networks.

- The weights/activation derivatives in different layers affect the backpropagated gradient in a multiplicative way.
  - With increasing depth this effect is magnified.
  - The partial derivatives can either increase or decrease with depth.

Example



- Neural network with one node per layer.
- Forward propagation multiplicatively depends on each weight and activation function evaluation.
- Backpropagated partial derivative get multiplied by weights and activation function derivatives.
- Unless the values are exactly one, the partial derivatives will either continuously increase (explode) or decrease (vanish).
- Hard to initialize weights exactly right.

Activation Function Propensity to Vanishing Gradients

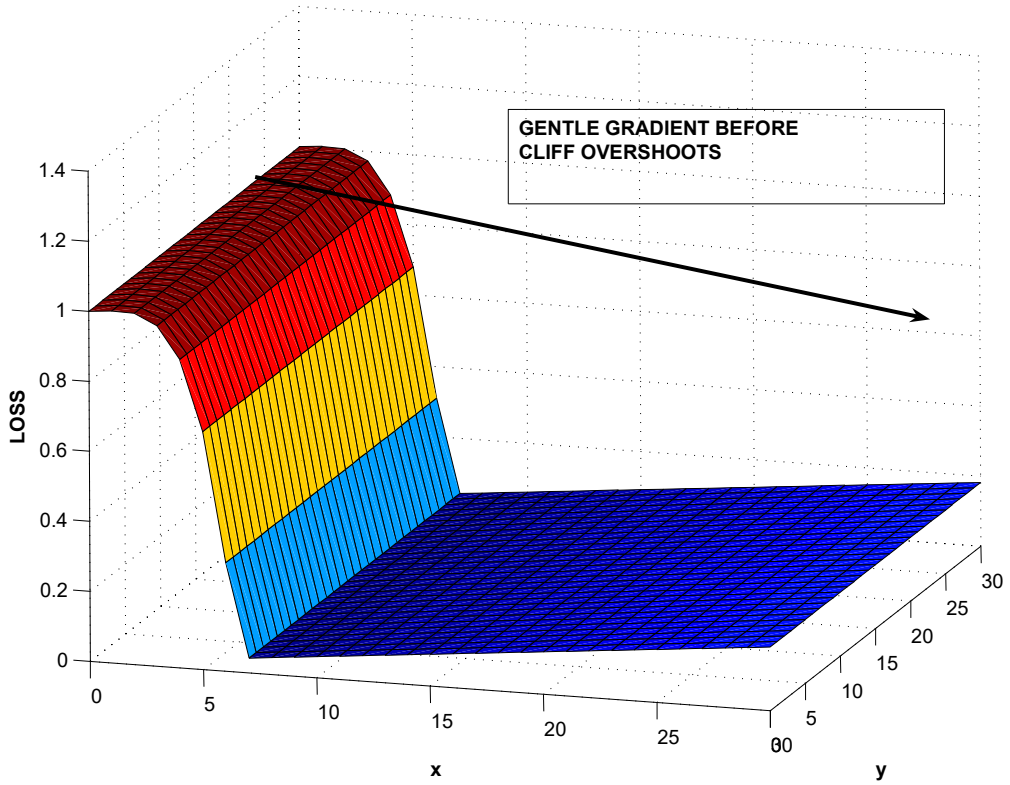
- Partial derivative of sigmoid with output $o \Rightarrow o(1-o)$.
  - Maximum value at $o = 0.5$ of 0.25.
  - For 10 layers, the activation function alone will multiply by less than $0.25^{10} \approx 10^{-6}$.
- At extremes of output values, the partial derivative is close to 0, which is called *saturation*.
- The tanh activation function with partial derivative $(1-o^2)$ has a maximum value of 1 at $o = 0$, but saturation will still cause problems.

Exploding Gradients

- Initializing weights to very large values to compensate for the activation functions can cause exploding gradients.

- Exploding gradients can also occur when weights across different layers are shared (e.g., recurrent neural networks).
  - The effect of a finite change in weight is extremely unpredictable across different layers.
  - Small *finite* change changes loss negligibly, but a slightly larger value might change loss drastically.

Cliffs



- Often occurs with the exploding gradient problem.

A Partial Fix to Vanishing Gradients

- The ReLU has linear activation for nonnegative values and otherwise sets outputs to 0.

- The ReLU has a partial derivative of 1 for nonnegative inputs.

- However, it can have a partial derivative of 0 in some cases and never get updated.

  – Neuron is permanently dead!

Leaky ReLU

- For negative inputs, the leaky ReLU can still propagate some gradient backwards.
  - At the reduced rate of $\alpha < 1$ times the learning case for nonnegative inputs:

$$\Phi(v) = \begin{cases} \alpha \cdot v & v \leq 0 \\ v & \text{otherwise} \end{cases} \tag{14}$$

- The value of α is a hyperparameter chosen by the user.

- The gains with the leaky ReLU are not guaranteed.

Maxout

- The activation used is $\max\{\overline{W_1} \cdot \overline{X}, \overline{W_2} \cdot \overline{X}\}$ with two coefficient vectors.

- One can view the maxout as a generalization of the ReLU.
  - The ReLU is obtained by setting one of the coefficient vectors to 0.
  - The leaky ReLU can also be simulated by setting the other coefficient vector to $\overline{W_2} = \alpha \overline{W_1}$.

- Main disadvantage is that it doubles the number of parameters.

Gradient Clipping for Exploding Gradients

- Try to make the different components of the partial derivatives more even.
  - *Value-based clipping:* All partial derivatives outside ranges are set to range boundaries.
  - *Norm-based clipping:* The entire gradient vector is normalized by the L_2-norm of the entire vector.
- One can achieve a better conditioning of the values, so that the updates from mini-batch to mini-batch are roughly similar.
- Prevents an anomalous gradient explosion during the course of training.

Other Comments on Vanishing and Exploding Gradients

- The methods discussed above are only partial fixes.

- Other fixes discussed in later lectures:
  - Stronger initializations with pretraining.
  - Second-order learning methods that make use of second-order derivatives (or *curvature* of the loss function).